

# The Projective Fiber Structure of $x + 1/x$ over Finite Fields

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## Abstract

Let  $q$  be an odd prime power and consider the rational map

$$\theta : \mathbf{P}^1(\mathbb{F}_q) \rightarrow \mathbf{P}^1(\mathbb{F}_q), \quad \theta(x) = x + \frac{1}{x},$$

with  $\theta(0) = \infty$  and  $\theta(\infty) = \infty$ . We prove an exact projective fiber law for  $\theta$ , derive the full indegree classification on its functional graph, compute the image cardinality, and obtain a clean algebraic constraint on genuine 2-cycles. We then sharpen this to a complete existence-and-uniqueness theorem for affine 2-cycles, and in the split case  $q \equiv 1 \pmod{4}$  we give an exact criterion for affine 3-cycles: they exist if and only if  $q \equiv 1 \pmod{36}$ . We also obtain an exact affine 4-cycle decomposition in the split case: one family is governed by  $q \equiv 1 \pmod{16}$  and a second family by  $q \equiv 1 \pmod{60}$ . These results give a self-contained structural core for the dynamics of  $x + 1/x$  over finite fields. We also explain why one should distinguish this map from the standard Lattès map attached directly to the CM endomorphism  $[1 + i]$  on the curve  $y^2 = x^3 + x$ .

*Formalization status.* The affine and projective fiber laws, the reciprocal-preimage lemma, the projective fiber above  $\infty$ , the non-existence of affine fixed points, the 2-cycle polynomial constraint, the 2-cycle characterization ( $x^2 = -1/2$ ), the affine image size  $(q + 1)/2$ , the no-preimage count  $(q - 1)/2$ , the projective image size  $(q + 3)/2$ , the full 3-cycle existence criterion (Theorem 16, including the Frobenius/cyclotomic argument via `AdjoinRoot`), and the split-case affine 4-cycle count (Theorem 19) have been verified in Lean 4 with Mathlib without any `sorry`. The remaining packaging gap is the indegree classification (Theorem 6) as a single final Lean statement.

## 1 Introduction

Let  $q$  be an odd prime power. The map

$$\theta(x) = x + \frac{1}{x} = \frac{x^2 + 1}{x}$$

defines a rational self-map of the projective line  $\mathbf{P}^1(\mathbb{F}_q)$  once we set

$$\theta(0) = \infty, \quad \theta(\infty) = \infty.$$

Its *functional graph* is the directed graph with vertex set  $\mathbf{P}^1(\mathbb{F}_q)$  and one edge

$$x \rightarrow \theta(x)$$

leaving each vertex.

The purpose of this note is modest and precise. We do not attempt a complete classification of all connected components of this graph. Instead, we isolate a theorem package that is already closed and robust:

- (i) the exact affine and projective fibers of  $\theta$ ;

- (ii) the full indegree classification on  $\mathbf{P}^1(\mathbb{F}_q)$ ;
- (iii) the image cardinality formula;
- (iv) the exact existence criterion for genuine affine 2-cycles;
- (v) an exact 3-cycle criterion in the split case  $q \equiv 1 \pmod{4}$ .
- (vi) an exact affine 4-cycle count in the split case  $q \equiv 1 \pmod{4}$ .

These results are sufficiently sharp to support a short note on their own, and they provide a reliable base layer for any future complete classification.

## 2 The Exact Fiber Law

**Definition 1.** *Let  $q$  be an odd prime power. Define*

$$\theta : \mathbf{P}^1(\mathbb{F}_q) \rightarrow \mathbf{P}^1(\mathbb{F}_q), \quad \theta(x) = x + \frac{1}{x}$$

*with  $\theta(0) = \infty$  and  $\theta(\infty) = \infty$ .*

**Proposition 2** (Affine fiber law). *Let  $a \in \mathbb{F}_q$  and let  $t \in \mathbb{F}_q^\times$ . Then*

$$\theta(t) = a \iff t^2 - at + 1 = 0.$$

*In particular, the affine preimages of  $a$  are exactly the roots in  $\mathbb{F}_q$  of the polynomial  $t^2 - at + 1$ .*

*Proof.* For  $t \neq 0$ ,

$$\theta(t) = a \iff t + \frac{1}{t} = a \iff t^2 - at + 1 = 0.$$

□

**Corollary 3** (Exact affine fiber structure). *Let  $a \in \mathbb{F}_q$  and let  $\Delta(a) = a^2 - 4$ . Then:*

- (i) *if  $\Delta(a)$  is a nonzero square in  $\mathbb{F}_q$ , then  $a$  has exactly two distinct affine preimages;*
- (ii) *if  $\Delta(a) = 0$ , equivalently  $a = \pm 2$ , then  $a$  has exactly one affine preimage, namely  $\pm 1$ ;*
- (iii) *if  $\Delta(a)$  is a nonsquare in  $\mathbb{F}_q$ , then  $a$  has no affine preimage.*

*Whenever two distinct affine preimages exist, they are reciprocal to one another.*

*Proof.* By Proposition 2, affine preimages are roots of a quadratic polynomial with discriminant  $\Delta(a) = a^2 - 4$ . The trichotomy follows from the usual root-count for quadratics over a field of odd characteristic. If  $r, s$  are the two roots, then Vieta's formula gives  $rs = 1$ , so  $s = r^{-1}$ . □

**Proposition 4** (Projective fiber above infinity). *The projective fiber above  $\infty$  is exactly*

$$\theta^{-1}(\infty) = \{0, \infty\}.$$

*Proof.* By definition,  $\theta(0) = \infty$  and  $\theta(\infty) = \infty$ . No nonzero affine point maps to  $\infty$ , because for  $t \in \mathbb{F}_q^\times$  the value  $\theta(t)$  is affine. □

**Corollary 5** (Projective fiber law). *For every affine value  $a \in \mathbb{F}_q$ , the projective fiber above  $a$  consists exactly of the nonzero roots of  $t^2 - at + 1 = 0$ , viewed as points of  $\mathbf{P}^1(\mathbb{F}_q)$ .*

### 3 Indegree Classification

**Theorem 6** (Indegree classification on  $\mathbf{P}^1(\mathbb{F}_q)$ ). *In the functional graph of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$ :*

- (i) *every vertex has indegree 0, 1, or 2;*
- (ii) *the vertex  $\infty$  has indegree 2;*
- (iii) *the only affine vertices of indegree 1 are 2 and  $-2$ ;*
- (iv) *an affine vertex  $a$  has indegree 2 if and only if  $a^2 - 4$  is a nonzero square in  $\mathbb{F}_q$ ;*
- (v) *an affine vertex  $a$  has indegree 0 if and only if  $a^2 - 4$  is a nonsquare in  $\mathbb{F}_q$ .*

*Proof.* Part (ii) is Proposition 4.

For affine vertices, Corollary 3 gives the complete root-count. Thus affine indegrees are exactly 0, 1, or 2, with indegree 1 occurring precisely when  $a = \pm 2$  and indegree 2 occurring precisely when  $a^2 - 4$  is a nonzero square. This proves all remaining parts.  $\square$

**Corollary 7** (Affine indegree counts). *Among the affine vertices of  $\mathbf{P}^1(\mathbb{F}_q)$ :*

- (i) *exactly 2 vertices have indegree 1;*
- (ii) *exactly  $\frac{q-3}{2}$  vertices have indegree 2;*
- (iii) *exactly  $\frac{q-1}{2}$  vertices have indegree 0.*

*Proof.* By Theorem 6, indegree-1 vertices are exactly  $\pm 2$ , so there are two of them.

The set of affine image points is the disjoint union of:

- (a) the two points  $\pm 2$ , each with one affine preimage;
- (b) the affine points with two affine preimages.

Every nonzero affine source point lies in exactly one affine fiber, and the only fibers of size 1 are those above  $\pm 2$ . Hence

$$q - 1 = 1 + 1 + 2N,$$

where  $N$  is the number of affine vertices of indegree 2. Therefore

$$N = \frac{q-3}{2}.$$

The remaining affine vertices have indegree 0, so their number is

$$q - 2 - \frac{q-3}{2} = \frac{q-1}{2}.$$

$\square$

### 4 Image-Size Formulas

**Theorem 8** (Affine image size). *The image of  $\theta$  on  $\mathbb{F}_q^\times$  has cardinality*

$$|\theta(\mathbb{F}_q^\times)| = \frac{q+1}{2}.$$

*Proof.* Each affine image point has either one or two affine preimages. By Corollary 7, exactly two image points have one preimage and exactly  $\frac{q-3}{2}$  image points have two preimages. Therefore the affine image contains

$$2 + \frac{q-3}{2} = \frac{q+1}{2}$$

points. □

**Corollary 9** (Projective image size). *The image of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$  has cardinality*

$$|\theta(\mathbf{P}^1(\mathbb{F}_q))| = \frac{q+3}{2}.$$

*Proof.* By Proposition 4, the value  $\infty$  belongs to the image. Together with Theorem 8, this gives

$$\begin{aligned} |\theta(\mathbf{P}^1(\mathbb{F}_q))| &= 1 + |\theta(\mathbb{F}_q^\times)| \\ &= 1 + \frac{q+1}{2} \\ &= \frac{q+3}{2}. \end{aligned}$$

□

## 5 A First Cycle Constraint

**Proposition 10** (No affine fixed point). *The point  $\infty$  is the unique fixed point of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$ .*

*Proof.* Certainly  $\theta(\infty) = \infty$ . If  $x \in \mathbb{F}_q^\times$  satisfied  $\theta(x) = x$ , then  $1/x = 0$ , impossible. Also  $\theta(0) = \infty \neq 0$ . □

**Theorem 11** (A polynomial constraint for genuine 2-cycles). *Let  $x \in \mathbb{F}_q^\times$  satisfy  $\theta(x) \neq 0$  and  $\theta(\theta(x)) = x$ . Then*

$$(2x^2 + 1)(x^2 + 1) = 0.$$

*Proof.* Set  $y = \theta(x) = x + x^{-1}$ . The relation  $\theta(y) = x$  gives

$$y + \frac{1}{y} = x.$$

Substituting  $y = x + x^{-1}$  and clearing denominators yields

$$(2x^2 + 1)(x^2 + 1) = 0.$$

This is a routine polynomial elimination. □

**Corollary 12** (Consequences for genuine 2-cycles). *Let  $x \in \mathbb{F}_q^\times$  satisfy  $\theta(x) \neq 0$  and  $\theta(\theta(x)) = x$ . Then either*

$$x^2 = -1 \quad \text{or} \quad x^2 = -\frac{1}{2}.$$

*If moreover  $x^2 = -1$ , then  $\theta(x) = 0$ , contradicting the hypothesis  $\theta(x) \neq 0$ . Hence every genuine affine 2-cycle must satisfy*

$$x^2 = -\frac{1}{2}.$$

*Proof.* The factorization in Theorem 11 implies the stated alternatives. If  $x^2 = -1$ , then  $x^{-1} = -x$ , so

$$\theta(x) = x + \frac{1}{x} = x - x = 0,$$

contrary to hypothesis. □

**Corollary 13** (Exact classification of affine 2-cycles). *A genuine affine 2-cycle exists if and only if  $-2$  is a square in  $\mathbb{F}_q$ . When it exists, it is unique and has the form*

$$x \longleftrightarrow -x, \quad x^2 = -\frac{1}{2}.$$

*Proof.* If  $x$  lies on a genuine affine 2-cycle, Corollary 12 gives  $x^2 = -1/2$ . Hence  $-2$  must be a square.

Conversely, suppose  $x^2 = -1/2$ . Then  $x \neq 0$  and

$$x \cdot (-2x) = -2x^2 = 1,$$

so  $x^{-1} = -2x$ . Therefore

$$\theta(x) = x + \frac{1}{x} = x - 2x = -x$$

and similarly  $\theta(-x) = x$ . Thus  $\{x, -x\}$  is a genuine affine 2-cycle.

Finally, the equation  $t^2 = -1/2$  has either no solution or exactly the two solutions  $\pm x$ , and these two points form the same 2-cycle. □

**Proposition 14** (Third iterate identity). *Assume  $\text{char } \mathbb{F}_q \neq 3$ . For every  $x \in \mathbb{F}_q^\times$ ,*

$$\theta^3(x) - x = \frac{3x^6 + 9x^4 + 6x^2 + 1}{x(x^2 + 1)(x^4 + 3x^2 + 1)}.$$

*Moreover, the numerator and denominator are coprime in  $\mathbb{F}_q[x]$ .*

*Proof.* A direct simplification of  $\theta^3(x) - x$  gives the stated rational function identity. Over  $\mathbb{Z}[x]$ , the numerator and denominator have greatest common divisor 1; reducing modulo any prime different from 3 preserves this coprimality. □

**Theorem 15** (Polynomial criterion for affine 3-cycles). *Assume  $\text{char } \mathbb{F}_q \neq 3$ . For  $x \in \mathbb{F}_q^\times$ , the following are equivalent:*

- (i)  $x$  lies on a genuine affine 3-cycle of  $\theta$ ;
- (ii)  $\theta^3(x) = x$ ;
- (iii)

$$3x^6 + 9x^4 + 6x^2 + 1 = 0.$$

*Equivalently, with  $y = x^2$ , genuine affine 3-cycles are governed by the cubic equation*

$$3y^3 + 9y^2 + 6y + 1 = 0.$$

*In particular,  $\theta$  has at most two affine 3-cycles.*

*Proof.* The equivalence of (ii) and (iii) follows from Proposition 14, because the numerator and denominator are coprime. Thus any affine solution of (iii) makes  $\theta^3(x)$  well defined and equal to  $x$ .

Now assume (ii). By Proposition 10,  $x$  does not have period 1. It also cannot have period 2, because  $\theta^2(x) = x$  would imply

$$\theta^3(x) = \theta(x),$$

and together with  $\theta^3(x) = x$  this gives  $\theta(x) = x$ , impossible. Hence the period of  $x$  is exactly 3, so (ii) implies (i). The converse is immediate.

The reformulation in terms of  $y = x^2$  is obtained by substitution. Since the sextic in (iii) has degree 6, there are at most six affine period-3 points, hence at most two affine 3-cycles.  $\square$

**Theorem 16** (Exact existence criterion for affine 3-cycles when  $q \equiv 1 \pmod{4}$ ). *Let  $q = p^n$  be an odd prime power with  $q \equiv 1 \pmod{4}$  and  $p \neq 3$ . Then the following are equivalent:*

- (i)  $\theta$  has a genuine affine 3-cycle on  $\mathbb{F}_q$ ;
- (ii) the polynomial  $3x^6 + 9x^4 + 6x^2 + 1$  has a root in  $\mathbb{F}_q$ ;
- (iii)  $q \equiv 1 \pmod{36}$ .

Whenever these conditions hold,  $\theta$  has exactly two affine 3-cycles.

*Proof.* Because  $q \equiv 1 \pmod{4}$ , there exists  $i \in \mathbb{F}_q$  with  $i^2 = -1$ . Set

$$f(x) = 3x^6 + 9x^4 + 6x^2 + 1, \quad m(u) = u^6 - 6u^4 + 9u^2 - 3.$$

Then for every  $x \in \mathbb{F}_q^\times$ ,

$$f(x) = -x^6 m(i/x),$$

so  $f$  has a root in  $\mathbb{F}_q$  if and only if  $m$  has a root in  $\mathbb{F}_q$ . By Theorem 15, this is equivalent to the existence of an affine 3-cycle.

*Backward direction* ( $q \equiv 1 \pmod{36} \implies m$  has a root). Since  $36 \mid (q-1)$  and  $\mathbb{F}_q^\times$  is cyclic of order  $q-1$ , there exists a primitive 36th root of unity  $\zeta \in \mathbb{F}_q$ . Set  $u = \zeta + \zeta^{-1}$ . Because  $\zeta^6$  is a primitive 6th root of unity, it satisfies  $\zeta^{12} - \zeta^6 + 1 = 0$ ; multiplying out gives

$$\zeta^6 \cdot m(u) = \zeta^{12} - \zeta^6 + 1 = 0,$$

so  $m(u) = 0$ .

*Forward direction* ( $m$  has a root  $\implies q \equiv 1 \pmod{36}$ ). Let  $u \in \mathbb{F}_q$  satisfy  $m(u) = 0$  and consider the quadratic  $g(t) = t^2 - ut + 1$ .

*Case 1:*  $g$  has a root  $\zeta \in \mathbb{F}_q$ . Then  $\zeta + \zeta^{-1} = u$  and  $m(u) = 0$ , so  $\zeta^{12} - \zeta^6 + 1 = 0$  by the same Chebyshev identity. One checks that  $\zeta^{36} = 1$  (squaring  $\zeta^{18} = -1$ ) and that no proper divisor of 36 annihilates  $\zeta$ : the relation  $\zeta^{18} = -1$  rules out every divisor of 18, while  $\zeta^{12} = 1$  would force  $\zeta^6 = 2$ , hence  $3 = 0$ , contradicting  $p \neq 3$ . Thus  $\zeta$  has order 36 in  $\mathbb{F}_q^\times$ , so  $36 \mid (q-1)$ .

*Case 2:*  $g$  is irreducible over  $\mathbb{F}_q$ . Let  $L = \mathbb{F}_q[t]/(g)$ , a degree-2 extension, and let  $\zeta$  be the image of  $t$  in  $L$ . The Frobenius  $\varphi: x \mapsto x^q$  satisfies  $\varphi(a) = a$  for  $a \in \mathbb{F}_q$ , so

$$\varphi(\zeta)^2 - u\varphi(\zeta) + 1 = (\zeta^2 - u\zeta + 1)^q = 0,$$

using  $(\alpha + \beta)^q = \alpha^q + \beta^q$  and  $(-1)^q = -1$  in characteristic  $p$ . Thus  $\varphi(\zeta)$  is also a root of  $g$ ; the two roots are  $\zeta$  and  $\zeta^{-1} = u - \zeta$ . Since  $g$  is irreducible over  $\mathbb{F}_q$ , the root  $\zeta$  is not Frobenius-fixed (all irreducible factors of  $X^q - X$  are linear), so  $\zeta^q = \zeta^{-1}$ . The same argument as Case 1 shows  $\zeta$  has order 36 in  $L^\times$ . Now  $\zeta^{q+1} = \zeta \cdot \zeta^{-1} = 1$ , so  $36 \mid (q+1)$ , giving  $q \equiv 35 \pmod{36}$ . But  $35 \equiv 3 \pmod{4}$ , contradicting  $q \equiv 1 \pmod{4}$ .

*Exact count.* The discriminant of  $f$  is  $-2^6 3^9$ , so in characteristic  $p \neq 2, 3$  the polynomial  $f$  has six distinct roots. Each such root has exact period 3 by Theorem 15. Hence there are exactly six affine period-3 points, that is, exactly two affine 3-cycles.  $\square$

**Proposition 17** (Fourth iterate identity). *For every odd prime power  $q$  and every  $x \in \mathbb{F}_q^\times$ ,*

$$\theta^4(x) - x = \frac{(2x^2 + 1)(2x^4 + 4x^2 + 1)(x^8 + 7x^6 + 14x^4 + 8x^2 + 1)}{x(x^2 + 1)(x^4 + 3x^2 + 1)(x^8 + 7x^6 + 13x^4 + 7x^2 + 1)}.$$

*Moreover, the numerator and denominator are coprime in  $\mathbb{Z}[x]$ .*

*Proof.* A direct simplification of  $\theta^4(x) - x$  gives the stated factorization. The numerator and denominator are coprime in  $\mathbb{Z}[x]$ , hence also after reduction modulo any odd prime.  $\square$

**Theorem 18** (Polynomial criterion for affine 4-cycles). *Let  $q$  be an odd prime power and let  $x \in \mathbb{F}_q^\times$ . Then  $x$  lies on a genuine affine 4-cycle of  $\theta$  if and only if*

$$(2x^4 + 4x^2 + 1)(x^8 + 7x^6 + 14x^4 + 8x^2 + 1) = 0.$$

*In particular,  $\theta$  has at most three affine 4-cycles.*

*Proof.* By Proposition 17, the condition  $\theta^4(x) = x$  is equivalent to

$$(2x^2 + 1)(2x^4 + 4x^2 + 1)(x^8 + 7x^6 + 14x^4 + 8x^2 + 1) = 0.$$

By Corollary 13, the factor  $2x^2 + 1$  cuts out exactly the affine period-2 points. Since  $\theta$  has no affine fixed point, an affine solution of  $\theta^4(x) = x$  has exact period 4 if and only if it is not of period 2. This proves the stated criterion.

The two exact-period factors have degrees 4 and 8, so there are at most 12 affine points of period 4, hence at most three affine 4-cycles.  $\square$

**Theorem 19** (Exact count of affine 4-cycles when  $q \equiv 1 \pmod{4}$ ). *Let  $q = p^n$  with  $p \equiv 1 \pmod{4}$ , and let  $N_4(q)$  denote the number of genuine affine 4-cycles of  $\theta$  on  $\mathbb{F}_q$ . Then*

$$N_4(q) = \begin{cases} 0, & q \not\equiv 1 \pmod{16} \text{ and } q \not\equiv 1 \pmod{60}, \\ 1, & q \equiv 1 \pmod{16} \text{ and } q \not\equiv 1 \pmod{60}, \\ 2, & q \not\equiv 1 \pmod{16} \text{ and } q \equiv 1 \pmod{60}, \\ 3, & q \equiv 1 \pmod{16} \text{ and } q \equiv 1 \pmod{60}. \end{cases}$$

*Equivalently,*

$$N_4(q) = \mathbf{1}_{q \equiv 1 \pmod{16}} + 2 \mathbf{1}_{q \equiv 1 \pmod{60}}.$$

*Proof.* Because  $p \equiv 1 \pmod{4}$ , we have  $q \equiv 1 \pmod{4}$ , so there exists  $i \in \mathbb{F}_q$  with  $i^2 = -1$ . Set

$$B(x) = 2x^4 + 4x^2 + 1, \quad C(x) = x^8 + 7x^6 + 14x^4 + 8x^2 + 1.$$

By Theorem 18, affine period-4 points are exactly the roots of  $B(x)C(x)$ .

For the quartic factor, write  $u = i/x$ . Then

$$x^4 B(x) = u^{-4}(u^4 - 4u^2 + 2),$$

so  $B(x) = 0$  if and only if  $u$  is a root of

$$m_{16}(u) = u^4 - 4u^2 + 2.$$

Now  $m_{16}$  is the minimal polynomial of

$$\alpha_{16} = \zeta_{16} + \zeta_{16}^{-1} = 2 \cos(\pi/8),$$

and  $\mathbb{Q}(\alpha_{16}) = \mathbb{Q}(\zeta_{16})^+$  is a Galois extension of degree 4. Hence  $m_{16}$  has a root in  $\mathbb{F}_q$  if and only if Frobenius at  $q$  acts trivially on  $\mathbb{Q}(\zeta_{16})^+$ , equivalently if and only if

$$q \equiv \pm 1 \pmod{16}.$$

Since  $q \equiv 1 \pmod{4}$ , this reduces to

$$q \equiv 1 \pmod{16}.$$

In that case  $m_{16}$  splits completely over  $\mathbb{F}_q$ , so  $B$  has exactly four roots in  $\mathbb{F}_q$ , producing exactly one affine 4-cycle.

For the octic factor, the same substitution gives

$$x^8 C(x) = u^{-8}(u^8 - 8u^6 + 14u^4 - 7u^2 + 1).$$

Set

$$m_{60}(u) = u^8 - 7u^6 + 14u^4 - 8u^2 + 1.$$

Then  $u^8 m_{60}(1/u) = u^8 - 8u^6 + 14u^4 - 7u^2 + 1$ , so  $C(x) = 0$  if and only if  $u = i/x$  is a root of the reciprocal polynomial of  $m_{60}$ . Now  $m_{60}$  is the minimal polynomial of

$$\alpha_{60} = \zeta_{60} + \zeta_{60}^{-1} = 2 \cos(\pi/30),$$

and  $\mathbb{Q}(\alpha_{60}) = \mathbb{Q}(\zeta_{60})^+$  is Galois of degree 8. Therefore  $m_{60}$  has a root in  $\mathbb{F}_q$  if and only if

$$q \equiv \pm 1 \pmod{60}.$$

Again  $q \equiv 1 \pmod{4}$  forces

$$q \equiv 1 \pmod{60}.$$

In that case  $m_{60}$  splits completely over  $\mathbb{F}_q$ , so  $C$  has exactly eight roots in  $\mathbb{F}_q$ , producing exactly two affine 4-cycles.

Finally,  $B$  and  $C$  are coprime in  $\mathbb{Z}[x]$ , so their root sets in  $\mathbb{F}_q$  are disjoint. Hence the total number of affine period-4 points is

$$4 \mathbf{1}_{q \equiv 1 \pmod{16}} + 8 \mathbf{1}_{q \equiv 1 \pmod{60}},$$

and division by 4 gives the stated formula for  $N_4(q)$ . □

## 6 A Corrected Isogeny Framing

**Remark 20** (Why this is not yet a complete classification). *The results above determine the exact fibers of  $\theta$ , the indegree distribution, the image size, the exact affine 2-cycles, the exact affine 3-cycle criterion in the split case  $q \equiv 1 \pmod{4}$ , and the exact affine 4-cycle count in the same split setting. They do not determine:*

- (i) *all cycle lengths;*
- (ii) *the number of cycles of each length;*
- (iii) *the full connected-component decomposition;*
- (iv) *the exact tree depth and branching profile in every component.*

*Thus they form a reliable theorem package, but not a complete classification of the functional graph.*



**Remark 21** (Corrected isogeny interpretation). *The map  $\theta(x) = x + 1/x$  arises naturally as the  $x$ -coordinate map of a degree-2 isogeny attached to the elliptic curve*

$$E : y^2 = x^3 + x.$$

*This gives a genuine CM background when  $p \equiv 1 \pmod{4}$ . However, one should not identify the full functional graph of  $\theta$  too quickly with the standard Lattès dynamics of the endomorphism  $[1 + i]$  without additional work. The exact relationship between the graph of  $\theta$  and the Gaussian-integer module picture is subtler than the naive slogan “ $\theta$  is the  $[1 + i]$ -Lattès map” suggests.*

## 7 A Proposed Research Roadmap

The problems below suggest individual targets, but the path toward a complete classification of the functional graph of  $\theta$  calls for a staged strategy rather than a direct assault. We record the following roadmap as a guide to future work.

**Stage 1. Structural foundation (done).** Establish the exact fiber law, indegree distribution, and image-size formulas. These results are closed, formally verified in Lean 4, and serve as the ground layer for everything that follows.

**Stage 2. Complete period-1 and period-2 classification (done).** Prove that  $\infty$  is the unique fixed point and that a genuine affine 2-cycle exists if and only if  $-2$  is a square in  $\mathbb{F}_q$ ; when it exists it is unique. This case is simple enough to close without auxiliary number-theoretic input.

**Stage 3. Complete period-3 classification (split case done).** The split case  $q \equiv 1 \pmod{4}$  is handled by Theorem 16 (criterion:  $q \equiv 1 \pmod{36}$ ); this case is now fully verified in Lean 4 with Mathlib, including the Frobenius/cyclotomic splitting argument via `AdjoinRoot`. Carry out the analogous analysis for the non-split case  $q \equiv 3 \pmod{4}$ , where the cyclotomic substitution via  $i$  is unavailable and a different approach is needed. The goal is an exact criterion covering all odd prime powers.

**Stage 4. Build a general exact-period elimination framework.** The iterate  $\theta^k(x) - x$  is a rational function whose numerator, after factoring out the contributions of all proper divisors of  $k$ , produces the “exact-period- $k$  polynomial”  $\Phi_k(x)$ . Develop the algebra of this factorization systematically: prove that the factors are coprime, determine their degrees, and identify the number-field structure (minimal polynomials of  $2 \cos(2\pi m/N)$  for suitable  $N$ ) governing their splitting over  $\mathbb{F}_q$ . The split-case analyses of periods 3 and 4 provide the first fully formalized instances of this framework in Lean 4.

**Stage 5. Complete period-4 and attack period-5.** The split case of period-4 is settled by Theorems 18 and 19, and this count is now fully formalized in Lean 4 with Mathlib, including the cyclotomic  $m_{16}/m_{60}$  splitting criteria and the characteristic-3 correction needed in the quartic/octic decomposition. The natural next step is period 5: identify the exact-period polynomial, determine the governing cyclotomic fields, and obtain a congruence criterion for the existence and count of affine 5-cycles.

**Stage 6. Distill the short-cycle number-field and Frobenius theory.** The pattern already visible in the completed split-case analyses of periods 3 and 4 is that exact-period polynomials have roots of the form  $2 \cos(\pi/m)$  for  $m$  determined by  $k$ , and splitting over  $\mathbb{F}_q$  is controlled by a Frobenius condition  $q \equiv \pm 1 \pmod{m}$ . Formulate and prove the general theorem relating period- $k$  cycle existence to the splitting of the relevant real cyclotomic field  $\mathbb{Q}(\zeta_m)^+$  at  $q$ .

**Stage 7. A unified short-cycle theorem up to a bound  $M$ .** Combine the results of Stages 2–6 into a single theorem: for every  $k \leq M$ , the exact number  $N_k(q)$  of genuine affine  $k$ -cycles is given by an explicit formula in terms of indicator functions  $\mathbf{1}_{q \equiv 1 \pmod{m_j}}$  for finitely many moduli  $m_j$  depending only on  $k$ .

**Stage 8. Tree structure and connected components.** With the cycle-vertex distribution known, study the in-trees attached to cycle vertices: determine the exact depth, the branching profile at each level, and hence the full description of every connected component as a directed cycle with rooted trees.

**Stage 9. Corrected CM and isogeny synthesis.** Once the full graph structure is explicit, return to the isogeny interpretation via  $E : y^2 = x^3 + x$  and the CM endomorphism  $[1 + i]$ . Identify precisely which features of the functional graph are predicted by the Gaussian-integer module  $E(\mathbb{F}_{p^n}) \cong \mathbb{Z}[i]/(\pi^n - 1)$ , and which require correction or supplementation.

**Stage 10. Full graph classification (terminal goal).** Assemble a complete, explicit description of the functional graph of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$  for every odd prime power  $q$ : all cycle lengths, all cycle counts, all tree depths and branching profiles, expressed in terms of arithmetic conditions on  $q$ .

## 8 Further Problems

**Problem 22** (Complete cycle classification). *Determine all cycle lengths occurring in the functional graph of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$ , and compute the exact number of cycles of each length.*

**Problem 23** (Component decomposition). *Describe every connected component of the functional graph of  $\theta$  on  $\mathbf{P}^1(\mathbb{F}_q)$  as a directed cycle with rooted trees attached to its cycle vertices.*

**Problem 24** (Tree depths and branching profiles). *For each cycle vertex  $c$ , determine the exact depth of the in-tree attached to  $c$  and the number of vertices at each level of that tree.*

**Problem 25** (Sharp criteria for short cycles). Solved for  $k = 2$  in Corollary 13, for  $k = 3$  in Theorem 16, and for  $k = 4$  in Theorem 19 (split case  $q \equiv 1 \pmod{4}$ ). *Find analogous exact existence-and-count criteria for genuine affine  $k$ -cycles for  $k \geq 5$ .*

**Problem 26** (Higher short cycles). *Theorems 15, 16, 18, and 19 settle affine 3- and 4-cycles in the split case. Push this to 5-cycles and beyond, preferably in a form that detects not only existence but also the exact number of short cycles.*

**Problem 27** (Lean formalization of remaining theorems). *The current Lean 4 formalization (Mathlib, no sorry) covers the fiber law, fixed-point results, 2-cycle characterization, image-size formulas, the third- and fourth-iterate identities (Propositions 14 and 17), the exact 3-cycle existence criterion (Theorem 16), and the split-case 4-cycle count formula (Theorem 19). For the latter, the Lean proof formalizes the  $m_{16}$  and  $m_{60}$  splitting criteria and treats the characteristic-3 quartic/octic overlap separately before assembling the final indicator-function formula. The main remaining theorem-level gap is the indegree classification (Theorem 6) in the same packaged form as the paper.*

**Problem 28** (Split CM classification). *When  $p \equiv 1 \pmod{4}$ , clarify the precise relationship between the functional graph of  $\theta$  and the Gaussian-integer module*

$$E(\mathbb{F}_{p^n}) \cong \mathbb{Z}[i]/(\pi^n - 1),$$

*and decide whether the full graph of  $\theta$  can ultimately be classified by a corrected CM/isogeny formalism.*